

Quantum-Enhanced Social Robotics: The QUADRI Project

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Abstract

This paper presents a review of the integration of quantum computing techniques into social robotics, focusing on the potential for enhancing robot adaptability, decision-making, and emotional intelligence. We analyze the current state of research, examining how quantum algorithms, such as Grover's algorithm, can be applied to improve human-robot interaction. The review, as a task of the QUADRI project (QUAntum-enhanced human-Robot Interaction: Pioneering Intelligent Social Robotics), provides a comprehensive overview of the opportunities and challenges in this emerging field, setting the foundation for future research and practical applications in domains such as mental health, education, and workplace stress management.

CCS Concepts

• **Human-centered computing** → **Interaction techniques**.

Keywords

social robotics, quantum computing, human robot interaction

ACM Reference Format:

Berardina Nadja De Carolis, Corrado Loglisci, Maria Grazia Miccoli, Giuseppe Palestra, and Sergio Violante. 2025. Quantum-Enhanced Social Robotics: The QUADRI Project. In *Adjunct Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP Adjunct '25)*, June 16–19, 2025, New York City, NY, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3708319.3735545>

1 Introduction

Quantum computing represents a significant shift from classical computing paradigms. The rapid evolution of quantum computing has introduced groundbreaking opportunities for various fields, with one of the most promising being its integration into social robotics. Social robotics aims to develop robots capable of natural, emotionally aware interactions with humans, an essential feature

for their application in daily life, healthcare, and educational settings [4, 5, 10, 27]. However, these systems often struggle with processing vast amounts of real-time data and adapting to dynamic, human-centered environments. Quantum computing, by leveraging quantum principles such as superposition and entanglement, offers solutions to these challenges, enabling robots to process information at unprecedented speeds and manage complex decision-making tasks more effectively. By analyzing a diverse range of studies, this study identifies promising applications of quantum computing in social robotics, including emotion recognition, path finding, and cognitive modeling. Moreover, it evaluates the feasibility of applying quantum algorithms such as Grover's search and quantum walk-based methods to improve robot adaptability and responsiveness to human emotional states. Despite the excitement surrounding these advancements, the integration of quantum computing into real-world robots still faces significant challenges, primarily related to hardware limitations and the complexity of quantum circuit implementation. Considering the importance that quantum computing could have for social robotics, it was decided to explore these topics further by conducting a state-of-the-art review, as detailed in the following sections of this paper. This review delves into the intersection of quantum computing and social robotics, highlighting the potential of Quantum Information Processing (QIP) to enhance robot behavior, decision-making, and emotional responsiveness. The study is based on an extensive review of more than 35 scientific articles published between 2021 and 2025, covering various aspects of this innovative integration. The primary aim is to examine current technological opportunities, the limitations of quantum computing in robotics, and potential future directions for the field. This paper provides a comprehensive analysis of these technological opportunities and barriers, offering original project proposals aimed at overcoming the limitations and fully harnessing the power of quantum-enhanced robotics.

2 The QUADRI project

The state-of-the-art review represents the first task of a research project called QUADRI. The QUADRI project (QUAntum-enhanced human-Robot Interaction: Pioneering Intelligent Social Robotics)



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ACM ISBN 979-8-4007-1399-6/25/06
<https://doi.org/10.1145/3708319.3735545>

pursues two main objectives aimed at extending an existing human-robot interaction framework: i) the application of quantum information processing algorithms that can be adapted and redefined in the context of social robotics technologies, with a direct impact on the user with whom the robot interacts; ii) the definition of new methodologies that make it possible to address issues traditionally studied using Reinforcement Learning techniques and agent-based systems through quantum search algorithms and quantum machine learning models. Analyzing and creating a natural and effective interaction with a social robot in complex contexts requires the use of models to interpret human behavior and, consequently, to plan the robot's response through personalized behaviors, which are then carried out via the robot's communicative actions. A social robot can be connected to a quantum simulator to enhance its processing and response capabilities in complex situations, thus becoming more social and more human-like. In the context of human-robot interaction, the predominant mode is verbal communication, where natural language text is extracted from the human's speech. This is the type of human stimulus on which this project proposal will focus. The robot processes stimuli through a decision-making system and acts accordingly in the external world, creating interactions based on the superposition of internal quantum states. This approach can be used to mimic human behavior, enabling the robot to carry out dialogues in response to stimuli. The integration of quantum computing with social robotics opens up new possibilities for the development of more sophisticated, autonomous, and intelligent robotic systems in complex environments. We believe that the QUADRI project will have a significant impact not only by advancing knowledge and technology, but also by improving people's quality of life and promoting a more equitable, inclusive, and sustainable society.

3 Related work

The papers examined in the current critical analysis can be grouped into five main categories, based on how quantum computing is applied within social robotics. These categories include:

- (1) reviews and theoretical foundations of quantum computing in social robotics;
- (2) cognitive models created through the integration of quantum technologies;
- (3) applications of quantum computing in emotion recognition and decision-making;
- (4) quantum approaches to pathfinding and navigation.

Each category highlights a specific aspect of how Quantum Information Processing can support more adaptive, responsive, and socially aware robotic systems. In the following sections, we will examine the most relevant contributions in each area, identifying both the progress made and the gaps still to be addressed.

3.1 Reviews and theoretical foundations of quantum computing in social robotics

Several review papers provide a comprehensive overview of quantum algorithms and technologies that are either already being applied or show promising potential within the context of social robotics. These works lay the theoretical foundation necessary to

understand how Quantum Information Processing might contribute to more adaptive and socially intelligent robotic systems.

The article *A Review on Quantum Search Algorithm* (2017) [8] offers a detailed analysis of Grover's algorithm and its many variations for searching in unstructured datasets. Although predating the most recent developments in quantum computing, this review remains highly relevant due to Grover's algorithm's continued prominence in quantum-enhanced decision-making and search tasks. In social robotics, such quantum search techniques could help optimize how a robot selects emotional responses or coping strategies from a large set of possible actions, especially when operating under time constraints or uncertainty.

The paper *Quantum Walk Computing: Theory, Implementation, and Application* (2024) [30] investigates graph-based quantum search techniques, particularly quantum walks. These methods are useful for solving pathfinding problems more efficiently than their classical counterparts. In a socially interactive setting, such as navigating a shared environment or dynamically adapting to human movement, the application of quantum walks could significantly improve a robot's spatial reasoning and decision-making capabilities.

A broader perspective is provided by *Advancements and Applications of Quantum Computing in Robotics* (2024) [11], which explores how quantum computing is being integrated into various aspects of robotics.

In *Quantum Robotics: A Review of Emerging Trends* (2024) [37], the authors focus on the current state of quantum hardware and software used in robotic systems. This review is particularly relevant for understanding the feasibility of deploying quantum technologies in real-world robots, including limitations in terms of integration, cost, and physical constraints. Such insights are crucial when envisioning future socially interactive robots that might include embedded quantum processors or cloud-accessible quantum resources.

The survey *A Survey on Quantum Reinforcement Learning* (2022) [26] examines how quantum techniques are being used to optimize reinforcement learning algorithms. Reinforcement learning is widely applied in robotics for training agents to make decisions through trial and error. Its quantum-enhanced versions show potential in accelerating learning processes and handling high-dimensional state spaces more effectively. These properties could be particularly useful in social robotics, where the robot must adapt its behavior to nuanced and changing human feedback.

Finally, *Quantum Operation of Affective Artificial Intelligence* (2023) [40] explores how quantum principles can support the development of affective AI systems that better mimic human decision-making processes. Across these reviews, the most prominent quantum technologies identified include *Grover's search algorithm*, *quantum walk-based graph search methods*, and the design of *quantum circuits* for learning and decision-making. Together, they represent a promising toolbox for enhancing the responsiveness, learning efficiency, and emotional intelligence of future social robots.

3.2 Cognitive models created through the integration of quantum technologies.

A growing body of research explores how quantum computing can contribute to the development of cognitive architectures in

artificial agents. These studies aim to model complex mental processes—such as perception, reasoning, creativity, and emotion—in ways that go beyond classical logic-based systems. In the context of social robotics, these quantum-inspired cognitive models offer new frameworks for designing robots capable of more human-like and context-sensitive behaviors.

One of the most significant contributions in this area is Quantum Affective Processes for Multidimensional Decision-Making (2022) [12], which introduces the theoretical framework Q-Coppelia. This model integrates cognitive and affective processing within a unified structure, inspired by quantum dynamics.

In Modeling Robotic Thinking and Creativity: A Classic–Quantum Dialogue (2024) [23], the authors propose a hybrid cognitive model grounded in the theories of D’Ariano and Faggin. This framework introduces quantum reasoning mechanisms into robotic cognition, aiming to simulate not just logical thinking, but also more abstract forms such as creativity and intuition.

The development of practical tools is addressed in The QUATRO Application Suite: Quantum Computing for Models of Human Cognition (2023) [29], which presents a collection of quantum algorithms specifically designed for modeling cognitive functions. The suite includes implementations that can be applied to simulate attention, memory, and decision-making processes, providing a concrete basis for experimentation in socially interactive systems.

Quantum Circuit Components for Cognitive Decision-Making (2023) [35] implements a simple quantum circuits that replicate human decision-making under uncertainty. These circuits process available information in a way that mimics how people often arrive at conclusions without clear-cut logic, which is a frequent challenge in social interaction scenarios.

Lastly, Unconventional Cognitive Intelligent Robotic Control: Quantum Soft Computing Approach in Human Being Emotion Estimation – QCOptKBTM Toolkit Application (2023) [34] presents a control model based on quantum soft computing. The system estimates human emotional states and adjusts robot behavior accordingly, integrating methods such as fuzzy logic and quantum walks.

The technologies most frequently used across these works include *fuzzy logic systems*, *quantum circuits*, *quantum walk algorithms*, and more specialized models such as *Multi-Particle-Multi-Well (MPMW)*, *Predator-Prey dynamics*, and the *Leaky Competing Accumulator (LCA) framework*. Each of these components contributes to building more nuanced and adaptable cognitive systems, paving the way for socially intelligent robots capable of reasoning, learning, and reacting in ways that mirror human thought.

3.3 Applications of quantum computing in emotion recognition and decision-making

Emotion recognition and decision-making are two of the most critical components in social robotics, as they directly impact a robot’s ability to interpret human affect and respond appropriately. In recent years, a wide range of studies have explored how quantum computing can enhance these processes, proposing both theoretical frameworks and practical implementations that offer clear benefits over classical models in terms of expressiveness, adaptability, and computational efficiency.

One of the earliest efforts in this direction is Quantum Motions and Emotions for a Humanoid Robot Actor (2021) [6], which explores the use of quantum circuits to generate expressive motion in humanoid robots. By leveraging the probabilistic and dynamic properties of quantum systems, the study demonstrates how a robot’s physical responses can be enriched to better reflect emotional states, a key asset in social interaction.

Decision-making under uncertainty is addressed in A Quantum-Like Model for Predicting Human Decisions in the Entangled Social Systems (2022) [24], which introduces a Quantum Bayesian Network capable of modeling entangled social dynamics. This approach has strong implications for human-robot collaboration, where robots must often anticipate and adapt to complex, context-sensitive choices.

The fusion of emotion recognition and quantum computing reaches another milestone in Quantum Computational Modeling for Affective Assessment in Virtual Reality Systems (2024) [2], where a Quantum Support Vector Machine (QSVM) is applied to classify emotional responses in augmented reality. Such systems are especially promising for applications in immersive therapy, education, and social training environments.

Advancements in quantum neural networks are exemplified by works like OQCNN: Optimal Quantum Convolutional Neural Network for Classification of Facial Expression (2023) [31] and A Deep Quantum Convolutional Neural Network Based Facial Expression Recognition for Mental Health Analysis (2024) [14], both of which utilize quantum-enhanced CNNs to improve the accuracy and efficiency of facial emotion recognition. These models outperform their classical counterparts in handling high-dimensional and noisy data, which are common challenges in real-world applications.

Multimodal emotion recognition is also an area where quantum approaches shine. A Quantum Probability Driven Framework for Joint Multi-Modal Sarcasm, Sentiment and Emotion Analysis (2023) [22] introduces the QUIET framework, capable of simultaneously processing text, video, and audio to distinguish subtle emotional cues, including sarcasm. Similarly, QMLSC: A quantum multimodal learning model for sentiment classification (2025) [20] and Hierarchical graph contrastive learning framework based on quantum neural networks for sentiment analysis (2025) [17] develop hybrid classical-quantum systems that merge various input streams into a coherent emotional profile.

The role of quantum optimization is evident in works like Efficient Facial Emotion Recognition Using an Optimized Deep Learning Model Based on Quantum Gazelle Optimization Algorithm (2024) [1], and Enhancing Feature Selection for Multi-Pose Facial Expression Recognition Using a Hybrid of Quantum-Inspired Firefly and Artificial Bee Colony Algorithm (2025) [28]. These studies illustrate how quantum-inspired methods can refine feature selection processes, thereby improving model performance and reducing computational costs.

Speech-based emotion recognition is also receiving attention, as seen in Advanced Speech Emotion Recognition Utilizing Optimized Equivariant Quantum Convolutional Neural Network for Accurate Emotional State Classification (2025) [7] and Decentralizing Feature Extraction with Quantum Convolutional Neural Network for Automatic Speech Recognition (2021) [38]. Both works show how quantum convolutional architectures can boost the sensitivity and

accuracy of vocal emotion classification, which is crucial for robots engaged in verbal interactions.

On a conceptual level, Conceptual Framework for Quantum Affective Computing and Its Use in Fusion of Multi-Robot Emotions (2021) [36] proposes a unique method of encoding and managing robotic emotional states using quantum superposition and unitary transformations. This opens new possibilities for emotional coordination among multiple robots, which could lead to more cohesive and natural group behaviors.

Quantum approaches have also been extended to applications like autonomous driving in Quantum Decision Making in Automatic Driving (2022) [33], and conversational emotion recognition in Quantum-Inspired Neural Network for Conversational Emotion Recognition (2021) [19], showing the versatility of these techniques across domains that demand real-time, adaptive affective processing.

Finally, video-based sentiment analysis benefits from frameworks like Quantum-Inspired Multimodal Fusion for Video Sentiment Analysis (2021) [18] and Quantum Cognitively Motivated Decision Fusion for Video Sentiment Analysis (2021) [9], which use principles such as entanglement and interference to model the interaction between visual, auditory, and textual modalities in sentiment recognition. and other papers in this context are given in the following references [21, 25, 28, 32, 39]

Across these studies, the most commonly adopted technologies include quantum circuits, quantum convolutional neural networks (QCNNs), quantum Bayesian networks, quantum-inspired recurrent networks (QRNNs), and quantum-enhanced optimization algorithms. These tools collectively provide a robust foundation for implementing emotion-aware and decision-capable robotic agents that can function effectively in dynamic and emotionally rich human environments.

3.4 Quantum approaches to pathfinding and navigation

Path planning is a fundamental component in mobile robotics, enabling robots to navigate through physical environments while avoiding obstacles and reaching designated goals. In recent years, quantum computing has opened new avenues for improving the efficiency and adaptability of pathfinding algorithms, especially in complex and dynamic environments.

One of the most promising examples [15] integrates quantum-enhanced reinforcement learning with fuzzy logic and artificial potential fields. This approach significantly improves the robot's ability to identify optimal paths in real time by leveraging the superposition and parallelism inherent in quantum systems.

In the context of swarm robotics, Quantum Planning for Swarm Robotics (2023) [3] introduces a quantum-inspired coordination mechanism that allows multiple robots to collaborate efficiently in identifying and following optimal routes toward shared objectives. This study demonstrates how quantum planning can enhance distributed decision-making and scalability in multi-agent systems.

Handling sparse-reward environments — scenarios where feedback is infrequent or delayed — is the focus of Quantum Exploration-Based Reinforcement Learning for Efficient Robot Path Planning in Sparse-Reward Environment (2024) [16]. The authors propose the

Quantum Exploration-Based Dreamer (QED), a novel algorithm that uses quantum-inspired exploration strategies to improve learning efficiency. This is especially beneficial in environments where traditional reinforcement learning struggles due to a lack of informative signals.

A more general framework is presented in Quantum Deep Reinforcement Learning for Robot Navigation Tasks (2024) [13], which combines parameterized quantum circuits (PQC) with classical reinforcement learning architectures. The resulting hybrid model is capable of faster convergence and more robust decision-making, especially in high-dimensional navigation problems.

Among the most used quantum technologies in these works are *quantum multi-stage Q-learning*, *parameterized quantum circuits (PQC)*, *Grover's algorithm*, *stochastic quantum random walkers*, and *quantum-enhanced exploration strategies*. These tools collectively provide innovative and efficient alternatives to traditional pathfinding approaches, making robots more autonomous and adaptive in navigating complex environments.

4 Conclusions

This section will discuss the gaps and opportunities that exist within scientific research that has just been analysed. Of course, each article has disadvantages relating to the particular application case, but a common shortcoming is that current hardware resources are not computationally suitable for quantum circuits or models consisting of many qubits. Instead, as far as common advantages are concerned, we find increased efficiency compared to classical algorithms, increased computational speed and better representation of human decision-making thanks to quantum techniques such as superposition or entanglement. The analysis showed that, to overcome the disadvantages of computational complexity, hybrid models are preferable for the time being, which manage and perceive the agent's environment in a classical way and process information in a quantum way (provided there are not too many of them or an efficient way to represent them in Hilbert space is found). Or they try to exploit quantum advantages by improving the classical algorithms by introducing small quantum circuits.

Acknowledgments

This work has been financed by the Research Program “National Quantum Science and Technology Institute” (PE0000023-NQSTI). It is supported within the framework of the National Recovery and Resilience Plan (NRRP), Mission 4 “Education and Research” – Component 2 “From Research to Business”, with explicit reference to funding provided by the European Union through the NextGenerationEU initiative.

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